

Type A - 2×5 pts

Q1. [5 pts] Let V and W be finite-dimensional vector spaces over a field F . Prove that V and W are isomorphic if and only if $\dim V = \dim W$.

Hints: An isomorphism $T : V \rightarrow W$ is an invertible linear transformation. What can you say about the null space and range of T ? Does the Rank-Nullity Theorem help in any way?

Rubric:

(a) Necessity ($V \cong W \implies \dim V = \dim W$)

- **1.0 pt:** Correctly identifies that an isomorphism $T : V \rightarrow W$ is a bijective linear transformation (specifically mentioning T is injective and surjective).
- **1.5 pt:** Applies the **Rank-Nullity Theorem** (0.5 pt for each):
 - T is injective $\implies \ker(T) = \{0\} \implies \text{nullity}(T) = 0$.
 - T is surjective $\implies \text{im}(T) = W \implies \text{rank}(T) = \dim W$.
 - Result: $\dim V = \text{nullity}(T) + \text{rank}(T) = 0 + \dim W$.

(b) Sufficiency ($\dim V = \dim W \implies V \cong W$)

- **1.5 pt:** let $n = \dim(V) \implies V \cong \mathbb{F}^n$ and $W \cong \mathbb{F}^n$.
- **1.0 pt:** since isomorphism is an equivalence relation, $V \cong W$

or

- **1.0 pt:** Construction of the Map: Let $n = \dim V = \dim W$. Choose bases $\mathcal{B}_V = \{v_1, \dots, v_n\}$ and $\mathcal{B}_W = \{w_1, \dots, w_n\}$. Define $T : V \rightarrow W$ such that $T(v_i) = w_i$.
- **1.5 pt:** Verification: Shows that this T is an isomorphism because it maps a basis to a basis, or explicitly shows T is linear and has an inverse.

(c) Common Deductions

- **-2.5 pts:** Only proved one direction of the "if and only if" statement.
- **-1.0 pt:** Failed to state that the constructed map T in Direction 2 is linear.

Q2. [5 pts] Suppose $T \in \mathcal{L}(\mathbb{C}^2)$ is defined by

$$T(w, z) = (-3z, w).$$

Find the eigenvalues of T and the corresponding eigenspaces. (Recall that the eigenspace corresponding to λ is the null space of $(T - \lambda I)$.)

Answer:

Let $(w, z) \neq (0, 0)$ be an eigenvector with eigenvalue λ . Then

$$T(w, z) = \lambda(w, z).$$

[1 mark]

Thus,

$$(-3z, w) = (\lambda w, \lambda z).$$

Equating components,

$$-3z = \lambda w \quad \text{and} \quad w = \lambda z.$$

[1 mark]

Substitute $w = \lambda z$:

$$-3z = \lambda^2 z.$$

Since $(w, z) \neq (0, 0)$, $z \neq 0$, hence

$$\lambda^2 = -3 \implies \lambda = \pm i\sqrt{3}.$$

[1 mark]

Eigenspaces:For $\lambda = i\sqrt{3}$:

$$(w, z) = z(i\sqrt{3}, 1), \quad E_{i\sqrt{3}} = \text{span}(i\sqrt{3}, 1). \quad [1 \text{ mark}]$$

For $\lambda = -i\sqrt{3}$:

$$(w, z) = z(-i\sqrt{3}, 1), \quad E_{-i\sqrt{3}} = \text{span}(-i\sqrt{3}, 1). \quad [1 \text{ mark}]$$

Q3. [5 pts] Recall the following proposition: Statements A to D below are equivalent.**A. T is diagonalisable****B. V has a basis consisting of eigenvectors of T** **C. There exist 1-dimensional subspaces $U_1, \dots, U_n$ of V , each invariant under T such that**

$$V = U_1 \oplus \dots \oplus U_n.$$

D. $\dim V = \dim E(\lambda_1, T) + \dots + \dim E(\lambda_m, T)$ **(where, recall $E(\lambda_i, T)$ is the eigenspace of the operator T corresponding to the eigenvalue λ_i).**Answer the following questions, based on this proposition. Consider the following matrix for T relative to the standard basis:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 12 \end{bmatrix}.$$

(a) [2.5 pts] Write down $U_1, \dots, U_3$, $E(1, T)$, $E(12, T)$ explicitly.**(b) [2.5 pts] Explicitly verify that all four conditions above hold (the first holds trivially, of course; do not assume the equivalence of A, B, C and D for this part).****Answer:**

Let

$$T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 12 \end{bmatrix}$$

be a linear operator on $V = \mathbb{R}^3$ or $\mathbb{V}^3$ or $\mathbb{F}^3$ or any abstract vectors.**(a) Finding U_1, U_2, U_3 , $E(1, T)$, and $E(12, T)$** The eigenvalues of T are given by its diagonal entries:

$$\lambda_1 = 1, \quad \lambda_2 = 12.$$

Eigenspace for $\lambda = 1$:

$$T - I = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix}.$$

Solving $(T - I)\mathbf{x} = 0$ gives $x_2 = 0$, $x_3 = 0$, with x_1 free. Hence,

$$E(1, T) = \text{span}\{(1, 0, 0)\}, \quad \dim E(1, T) = 1.$$

Eigenspace for $\lambda = 12$:

$$T - 12I = \begin{bmatrix} -11 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Solving $(T - 12I)\mathbf{x} = 0$ gives $x_1 = 0$, with x_2, x_3 free. Hence,

$$E(12, T) = \text{span}\{(0, 1, 0), (0, 0, 1)\}, \quad \dim E(12, T) = 2.$$

One-dimensional invariant subspaces:

$$U_1 = \text{span}\{(1, 0, 0)\}, \quad U_2 = \text{span}\{(0, 1, 0)\}, \quad U_3 = \text{span}\{(0, 0, 1)\}.$$

(b) Verification of conditions A, B, C, and D

A. T is diagonalisable.

Since T is a diagonal matrix, it is diagonalisable.

B. V has a basis consisting of eigenvectors of T .

The vectors $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ are eigenvectors of T corresponding to eigenvalues 1 and 12. These vectors form the standard basis of $\mathbb{R}^3$, hence a basis of eigenvectors exists.

C. $V = U_1 \oplus U_2 \oplus U_3$ with each U_i invariant under T .

We verify invariance:

$$T(1, 0, 0) = (1, 0, 0) \in U_1,$$

$$T(0, 1, 0) = (0, 12, 0) \in U_2,$$

$$T(0, 0, 1) = (0, 0, 12) \in U_3.$$

Thus, each U_i is invariant under T .

Further, the vectors $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ are linearly independent and span V . Hence,

$$V = U_1 \oplus U_2 \oplus U_3.$$

D. $\dim V = \sum \dim E(\lambda_i, T)$.

$$\dim V = 3,$$

$$\dim E(1, T) = 1, \quad \dim E(12, T) = 2.$$

$$\dim E(1, T) + \dim E(12, T) = 1 + 2 = 3 = \dim V.$$

Thus, all four conditions A, B, C, and D are verified explicitly.

Rubric (Total: 5 marks)

(a) [2.5 marks]

- Correct computation of $E(1, T)$: 0.75
- Correct computation of $E(12, T)$: 0.75
- Correct identification of U_1, U_2, U_3 : 1.0

(b) [2.5 marks]

- A: Correct justification of diagonalizability: 0.25
- B: Basis of eigenvectors clearly identified: 0.75
- C: Invariance and direct sum properly shown: 1.0
- D: Dimension equality correctly verified: 0.5

Type B - 1×8 pts

Q4. Let V be an n -dimensional vector space over F and let $\alpha_1, \dots, \alpha_n$ be a basis for V . Similarly, let W be an m -dimensional vector space over F and let $\beta_1, \dots, \beta_m$ be a basis for W . Let $\mathcal{L}(V, W)$ denote the set of linear operators from V to W . This question asks you to fill in some steps in the proof that the linear operators $E^{p,q} \in \mathcal{L}(V, W)$ defined as

$$E^{p,q}(\alpha_i) := \delta_{iq}\beta_p = \begin{cases} \beta_p, & \text{if } i = q, \\ 0, & \text{if } i \neq q \end{cases}$$

is a basis for $\mathcal{L}(V, W)$ where $p \in \{1, \dots, m\}$, $q \in \{1, \dots, n\}$.

(a) [4 pts] Show that if a linear operator $T \in \mathcal{L}(V, W)$ acts as

$$T\alpha_j = \sum_{p \in \{1, \dots, m\}} A_{pj}\beta_p,$$

then one can write

$$T = \sum_{p \in \{1, \dots, m\}, q \in \{1, \dots, n\}} A_{pq}E^{pq}.$$

Hint: Simply apply and see how α_i is transformed by $\sum_{pq} A_{pq}E^{pq}$.

(b) [4 pts] Let

$$U := \sum_{p \in \{1, \dots, m\}, q \in \{1, \dots, n\}} A_{pq}E^{pq}$$

be the zero transformation, i.e. $U(\alpha_i) = 0$ for all α_i . Using this, prove that $A_{pj} = 0$ for every p and j . *Hint:* It may help to show $\sum_p A_{pj}\beta_p = 0$ and use the fact that $\{\beta_j\}_j$ is a basis.

Answer:

(a) Consider $T \in \mathcal{L}(V, W)$ such that T acts

$$T\alpha_j = \sum_{p \in \{1, \dots, m\}} A_{pj}\beta_p$$

$\forall j \in \{1, \dots, n\}$ and α_j is a basis vector for V .

Now, from a known result, such a linear transformation is unique, hence it suffices to show that T can be expressed as a linear combination of the above Linear Transformations E^{pq} for simply the basis vectors.

Consider some arbitrary $j \in \{1, \dots, n\}$. We observe that

$$T\alpha_j = \sum_{p \in \{1, \dots, m\}} A_{pj} \left(E^{pj}(\alpha_j) + \sum_{q \in \{1, \dots, n\}/\{j\}} 0 \right)$$

But the above 0 can be written as $mE^{pq}(\alpha_j) \forall q \in \{1, \dots, n\}/\{j\}$ for any scalar m as per the definitions of the linear transformations E^{pq} . Therefore, we get

$$\begin{aligned} T\alpha_j &= \sum_{p \in \{1, \dots, m\}} A_{pj}E^{pj}(\alpha_j) + \left(\sum_{q \in \{1, \dots, n\}/\{j\}} A_{pq}E^{pq}(\alpha_j) \right) = \sum_{p \in \{1, \dots, m\}} \sum_{q \in \{1, \dots, n\}} A_{pq}E^{pq}(\alpha_j) \\ &= \sum_{p \in \{1, \dots, m\}; q \in \{1, \dots, n\}} A_{pq}E^{pq}(\alpha_j) \end{aligned}$$

The above holds for **all** $j \in \{1, \dots, n\}$. Thus, by the uniqueness of the Linear transformation enforced by its action on the basis vectors above, we can claim that

$$T = \sum_{p \in \{1, \dots, m\}; q \in \{1, \dots, n\}} A_{pq}E^{pq}$$

hence proved.

(b) Now we need to prove the linear independence of each of these transformations. So suppose that some linear combination of the transformations

$$\sum_{p \in \{1, \dots, m\}; q \in \{1, \dots, n\}} c_{pq}E^{pq} = 0$$

for some scalars c_{pq} . Here 0 represents the Null Transformation. Since, $0\alpha = 0$ for any $\alpha \in V$ the above also holds for all the basis vectors of V , i.e. $\alpha_i \forall i \in \{1, \dots, n\}$. Therefore for any arbitrary i we obtain

$$\begin{aligned} 0\alpha_i &= \sum_{p \in \{1, \dots, m\}; q \in \{1, \dots, n\}} c_{pq} E^{pq}(\alpha_i) \\ &= \sum_{p \in \{1, \dots, m\}} c_{pi} E^{pi}(\alpha_i) = \sum_{p \in \{1, \dots, m\}} c_{pi} \beta_p = 0_W \end{aligned}$$

Now, since $\{\beta_p\}_{p \in \{1, \dots, n\}}$ represent the basis, a linear combination is the zero vector iff each coefficient is 0. Therefore we must have $c_{pi} = 0 \forall p \in \{1, \dots, n\}$. This is true for any arbitrarily chosen i . So this must extend for all i 's. Hence we finally obtain that $c_{pq} = 0 \forall (p, q) \in \{1, \dots, m\} \times \{1, \dots, n\}$. Hence, E^{pq} are linearly independent. Hence proved.

Marking Scheme:

- **Part a (4m):**
 - Showing that the above linear combination acts the same as T on any arbitrarily chosen basis vector of V **1m**
 - Using the definition of E^{pq} appropriately. **1m**
 - How the single summation changes to a double summation over p, q . **1m**
 - Finally claiming that the above holds **for all** vectors in V , via an already proven result. **1m**
 - **Part b (4m):**
 - Any valid method which boils down to a linear combination of independent vectors in W . **2m**
 - Using the linear independence of the basis of W . **2m**
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Q5. Let V be an n (finite) dimensional complex vector space where $n > 0$. Suppose $T \in \mathcal{L}(V)$ is a linear operator. Show the following:

- (a) [2 pts] Let $v \in V$ with $v \neq 0$. Consider $v, Tv, T^2v, \dots, T^nv$. Show that they are linearly dependent.
 (b) [4 pts] Let $a_0, \dots, a_n$ be complex numbers (not all zero) satisfying

$$a_0v + a_1Tv + \dots + a_nT^nv = 0.$$

Suppose the following holds for any non-constant polynomial:

There exist complex numbers $c \neq 0$, $\lambda_1, \dots, \lambda_m$ and an integer $m \geq 1$, satisfying

$$a_0 + a_1z + \dots + a_nz^n = c(z - \lambda_1) \cdots (z - \lambda_m)$$

for all complex numbers z where $m \leq n$.

Show that

$$c(T - \lambda_1 I) \cdots (T - \lambda_m I)v = 0.$$

Hint: Can $a_1, \dots, a_n$ be all zero? Do you need this in your argument?

- (c) [2 pts] Given the equation above holds, prove that there exists at least one index j , such that the null space of $T - \lambda_j I$ has at least one non-zero vector.

Answer:

Let V be a complex vector space with $\dim(V) = n$, and let $T \in \mathcal{L}(V)$ be a linear operator.

- (a) Let $v \in V$ with $v \neq 0$. Then the $(n + 1)$ vectors

$$v, Tv, T^2v, \dots, T^nv$$

are not linearly independent.

Hence,

$$v, Tv, \dots, T^nv \text{ are linearly dependent.}$$

- (b) From part (a), since $v, Tv, \dots, T^nv$ are linearly dependent, there exist complex numbers $a_0, \dots, a_n$, not all zero, such that

$$0 = a_0v + a_1Tv + \dots + a_nT^nv.$$

Here $a_1, \dots, a_n$ cannot all be 0, because otherwise the equation above would become $0 = a_0v$, which would force a_0 also to be 0. Hence, the polynomial is non-constant. We then have

$$a_0I(v) + a_1T(v) + \dots + a_nT^n(v) = 0.$$

As T is linear, we can write

$$(a_0I + a_1T + \dots + a_nT^n)v = 0.$$

It is given that there exist complex numbers $c \neq 0$, $\lambda_1, \dots, \lambda_m$ and an integer $m \geq 1$, satisfying

$$a_0 + a_1z + \dots + a_nz^n = c(z - \lambda_1) \cdots (z - \lambda_m)$$

for all complex numbers z where $m \leq n$. Now,

$$\begin{aligned} & (a_0I + \dots + a_nT^n)v = 0 \\ \implies & c(T - \lambda_1 I) \cdots (T - \lambda_m I)v = 0 \end{aligned}$$

- (c) From (b),

$$c(T - \lambda_1 I) \cdots (T - \lambda_m I)v = 0.$$

Since $v \neq 0$, $T - \lambda_j I$ is not injective for at least one j . Therefore,

$$\ker(T - \lambda_j I) \neq 0.$$

Thus, there exists at least one index j such that the null space of $T - \lambda_j I$ contains at least one non-zero vector.

Rubric:

Part (a):

- [1 Mark] For stating there are $n + 1$ vectors in the set and $\dim(V) = n$.
- [1 Mark] For concluding that these vectors are linearly dependent.

Part (b):

- [1 Mark] Using $v, Tv, \dots, T^n v$ are linearly dependent and write it.
- [1 Mark] Writing the fact that $a_1, \dots, a_n$ cannot all be 0.
- [1 Mark] Writing the operator form and linearity of T to get the expression.
- [1 Mark] Using the factorization and writing the conclusion.

Part (c):

- [1 Mark] Mention why $T - \lambda_j I$ is not injective for at least one j .
 - [1 Mark] Writing the conclusion with the reason.
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